

Finite-Blocklength Latency Minimization for Uplink and Downlink Communications: Method Results: Interference

July 6, 2026

Abstract

This document collects the exported interference figures for uplink and downlink under payload completion. Throughout, the *Training Only Method* is used as the benchmark for the *Training + Testing Method*. The uplink figures mostly show moderate interference reshaping inside a similar operating regime. The downlink figures are more structural, especially when users are removed from the active set and their traces drop toward -300 dB.

1 Scope and Naming

This document is intentionally interference-only. The terminology is aligned with the companion theory report:

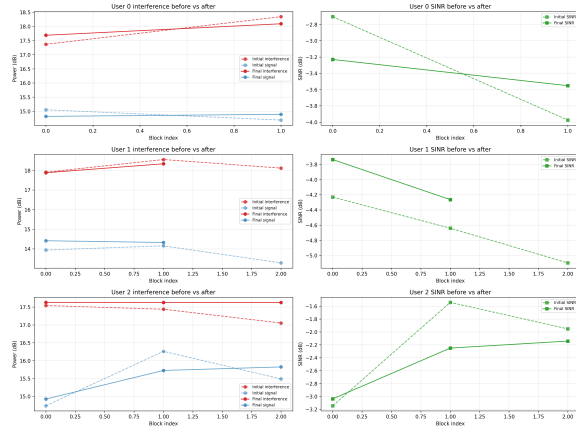
- **Training Only Method:** direct online optimization during scheduling,
- **Training + Testing Method:** a separate training stage followed by testing-stage inference,
- **Payload completion:** the scenario where each user continues until its remaining payload is exhausted.

2 Uplink Payload Completion

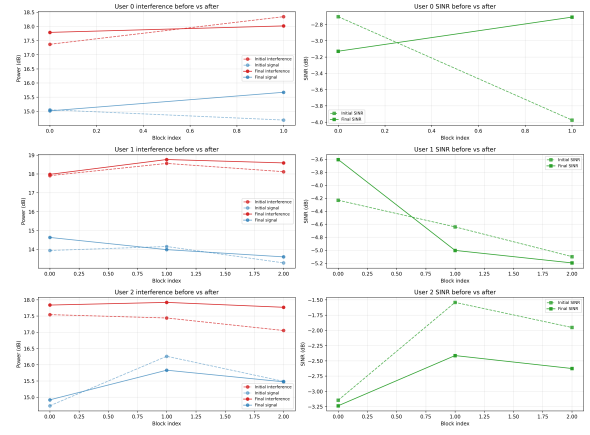
The uplink payload-completion interference picture stays fairly close across the two methods. The benchmark does improve final SINR for users 1 and 2, but the overall difference remains moderate rather than dramatic.

Table 1: Uplink payload-completion per-user SINR summary

User	Training Only init/final	Training + Testing init/final	Reading
0	$-3.359/-3.396$ dB	$-3.359/-2.908$ dB	Training + Testing is better on this user, but not enough to change the system outcome.
1	$-4.649/-4.006$ dB	$-4.649/-4.592$ dB	Benchmark has the clearest SINR gain here.
2	$-2.173/-2.459$ dB	$-2.173/-2.743$ dB	Benchmark ends less negative.

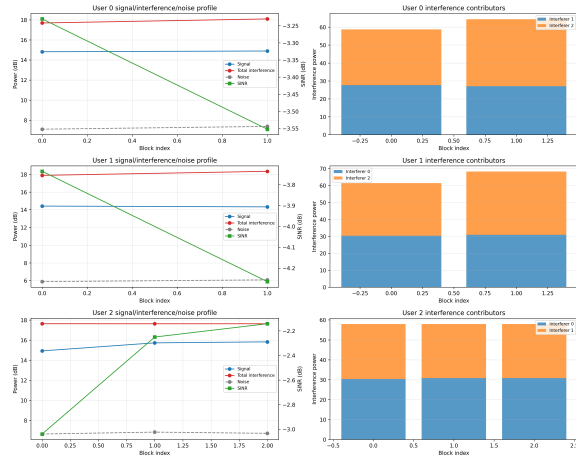


(a) Training Only Method

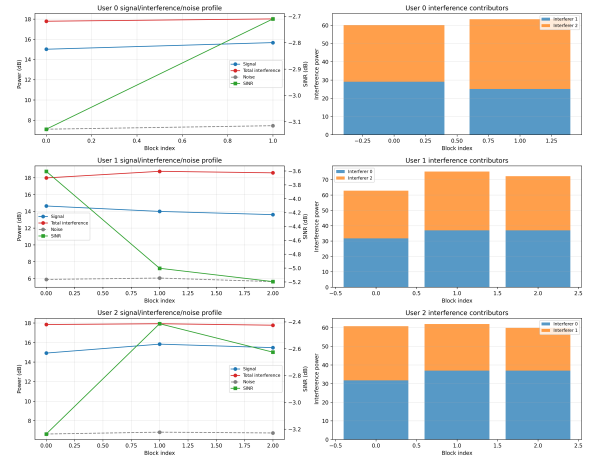


(b) Training + Testing Method

Figure 1: Uplink payload-completion per-user interference before/after comparison.

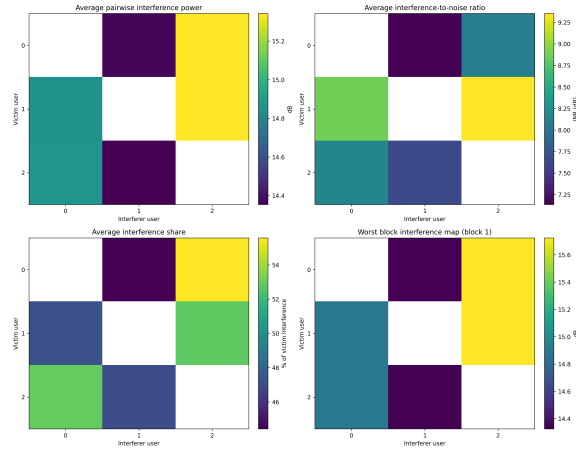


(a) Training Only Method

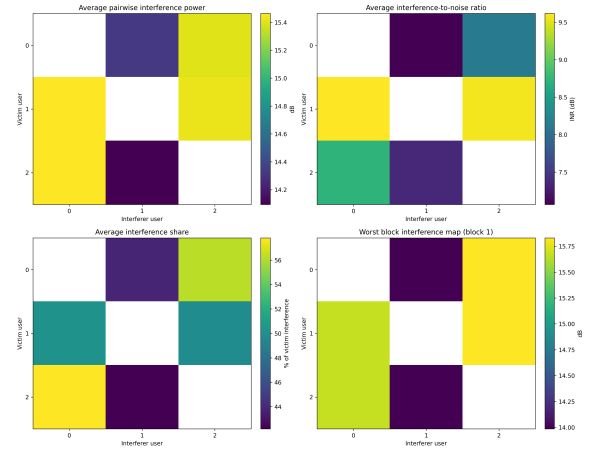


(b) Training + Testing Method

Figure 2: Uplink payload-completion per-user interference profiles.



(a) Training Only Method



(b) Training + Testing Method

Figure 3: Uplink payload-completion interference heatmaps.

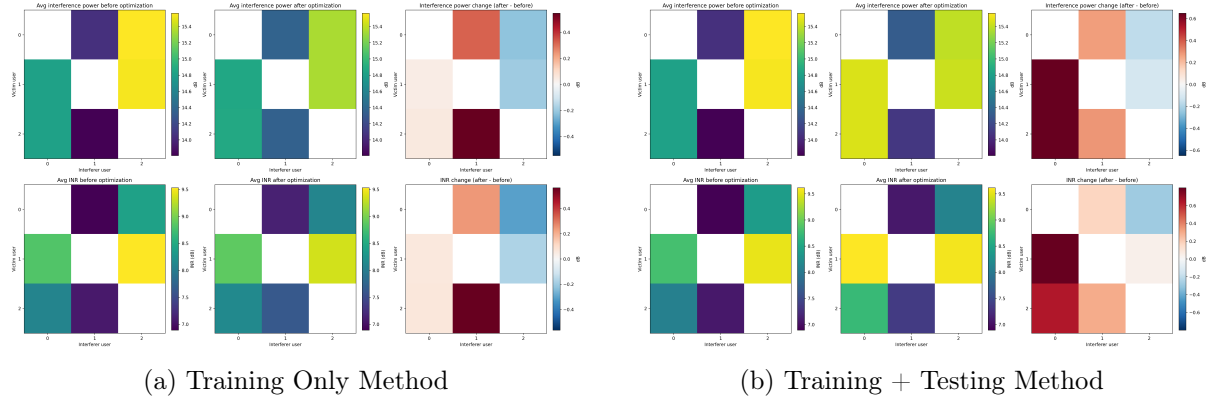


Figure 4: Uplink payload-completion before/after heatmap comparison.

3 Downlink Payload Completion

The downlink payload-completion interference results are more structural. The benchmark clears users from the active set more decisively, and that shows up in both the -300 dB inactive segments and the shorter schedule tail.

Table 2: Downlink payload-completion per-user served-block SINR summary

User	Training Only init/final	Training + Testing init/final	Reading
0	$-3.003/18.920$ dB	$-3.003/14.017$ dB	Benchmark is much stronger after skipped-block concentration.
1	$-2.774/10.052$ dB	$-2.774/13.141$ dB	Training + Testing is stronger on this easier user.
2	$-2.764/12.894$ dB	$-2.764/11.416$ dB	Benchmark is stronger and finishes earlier.

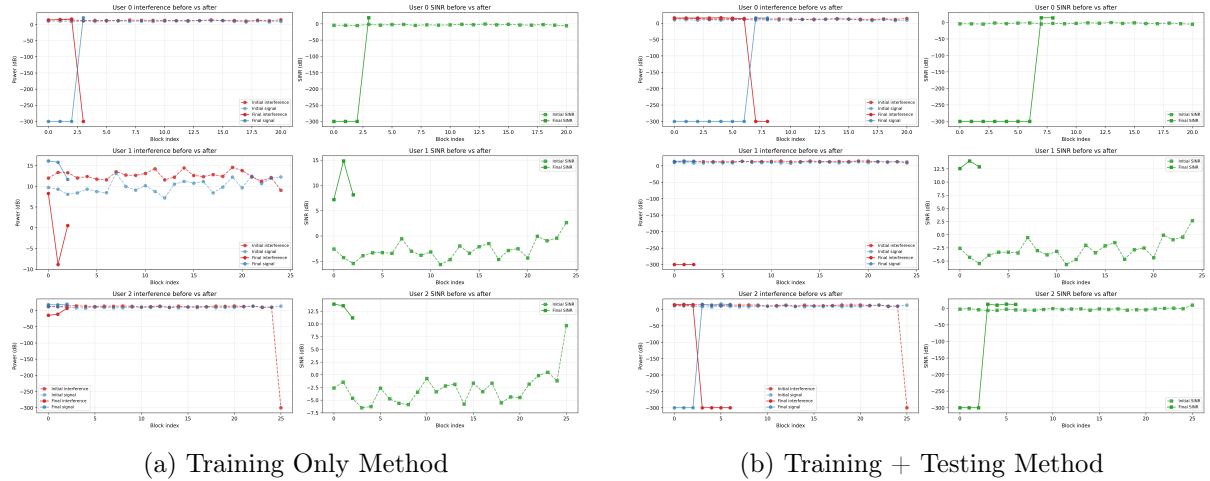
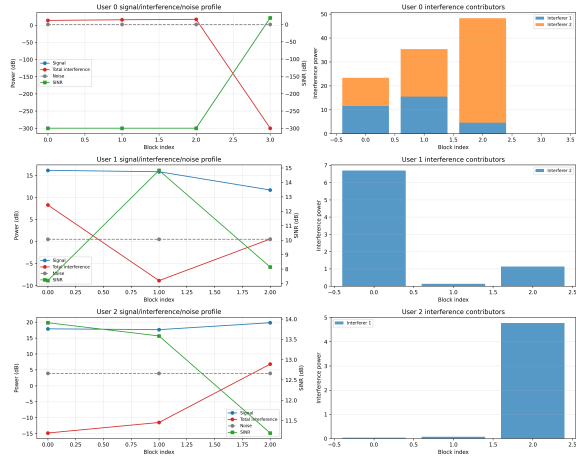
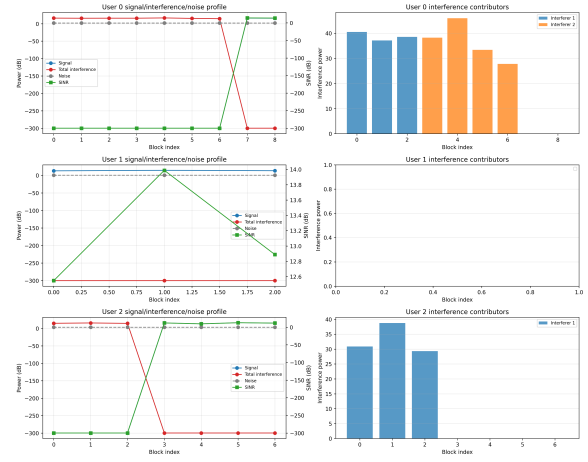


Figure 5: Downlink payload-completion per-user interference before/after comparison.

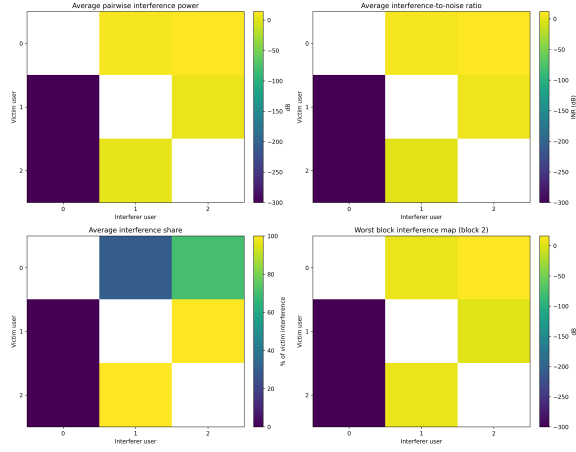


(a) Training Only Method

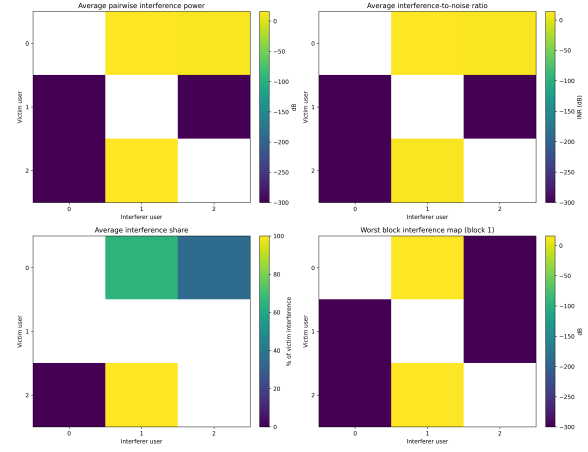


(b) Training + Testing Method

Figure 6: Downlink payload-completion per-user interference profiles.

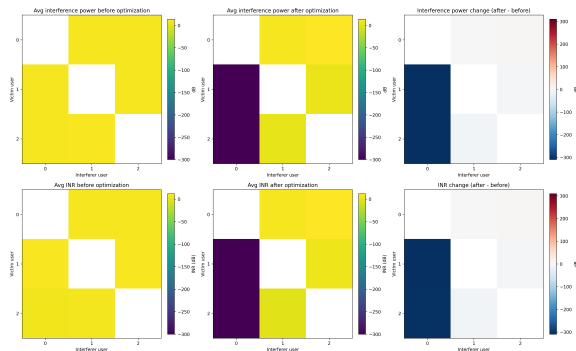


(a) Training Only Method

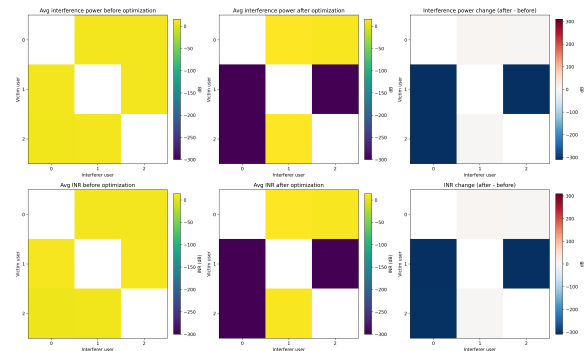


(b) Training + Testing Method

Figure 7: Downlink payload-completion interference heatmaps.



(a) Training Only Method



(b) Training + Testing Method

Figure 8: Downlink payload-completion before/after heatmap comparison.

4 Interference Takeaways

Across all exported figures, the uplink remains the closer match between methods. Its interference figures mostly show small SINR redistributions inside a similar operating range. The downlink is more revealing. In payload completion, the benchmark concentrates service more decisively on the currently useful users.